

Database Technologies – Assignment 8

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD1_Rosh_98811> SELECT * FROM ORDERS WHERE CNUM = ( SELECT CNUM FROM CUSTOMERS WHERE CNAME = 'CISNEROS' );
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3006	1098.16	1990-10-03	2008	1007

2 rows in set (0.045 sec)

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD1_Rosh_98811> SELECT o.amt, c.cname, c.rating FROM Orders o JOIN Customers c ON o.cnum = c.cnum WHERE o.amt > (SELECT AVG(amt) FROM Orders);
```

amt	cname	rating
5160.45	Liu	200
4723.00	Clemens	100
9891.88	Clemens	100

3 rows in set (0.040 sec)

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD1_Rosh_98811> SELECT snum, SUM(amt) AS total_amount FROM Orders GROUP BY snum HAVING SUM(amt) > (SELECT MAX(amt) FROM Orders);
```

snum	total_amount
1001	15382.07

1 row in set (0.018 sec)

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD1_Rosh_98811> SELECT * FROM Customers WHERE rating >= ANY (SELECT rating FROM Customers WHERE snum = (SELECT snum FROM Salespeople WHERE sname = 'Serres'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.031 sec)

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD1_Rosh_98811> SELECT * FROM Salespeople s WHERE s.city!= ALL (SELECT c.city FROM Customers c WHERE c.snum = s.snum);
```

snum	sname	city	comm
1004	Motika	London	0.11
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

3 rows in set (0.129 sec)

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD1_Rosh_98811> SELECT * FROM Orders WHERE amt > ANY (SELECT amt FROM Orders WHERE cnum IN (SELECT cnum FROM Customers WHERE city = 'London'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.067 sec)

7. Extract all the orders of Motika.

```
KD1_Rosh_98811> SELECT * FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.030 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD1_Rosh_98811> SELECT * FROM Orders WHERE snum IN (SELECT snum FROM Customers WHERE city = 'London');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.017 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD1_Rosh_98811> SELECT snum, sname FROM Salespeople WHERE snum IN (SELECT snum FROM Customers GROUP BY snum HAVING COUNT(*) > 1);
+-----+-----+
| snum | sname |
+-----+-----+
| 1001 | Peel  |
| 1002 | Serres |
+-----+-----+
2 rows in set (0.057 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD1_Rosh_98811> SELECT s.snum, s.sname, s.city, (SELECT COUNT(*) FROM Customers
c WHERE c.snum = s.snum) AS customer_count FROM Salespeople s WHERE s.snum IN
(SELECT snum FROM Customers GROUP BY snum HAVING COUNT(*) > 1);
+-----+-----+-----+-----+
| snum | sname | city   | customer_count |
+-----+-----+-----+-----+
| 1001 | Peel  | London | 2 |
| 1002 | Serres | San Jose | 2 |
+-----+-----+-----+-----+
2 rows in set (0.039 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD1_Rosh_98811> SELECT * FROM Customers WHERE rating > ANY (SELECT rating FROM Customers WHERE city = 'Rome');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.489 sec)

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD1_Rosh_98811> SELECT * FROM Orders WHERE amt > ANY (SELECT amt FROM Orders WHERE odate = '1990-10-04');
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.760 sec)

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD1_Rosh_98811> SELECT * FROM Orders WHERE amt < ANY (SELECT amt FROM Orders WHERE cnum IN (SELECT cnum FROM Customers WHERE city = 'San Jose'));
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.221 sec)

14. Select those customers whose rating are higher than every customer in London.

```
KD1_Rosh_98811> SELECT * FROM Customers WHERE rating > ALL (SELECT rating FROM Customers WHERE city = 'London');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.020 sec)